MATH HW

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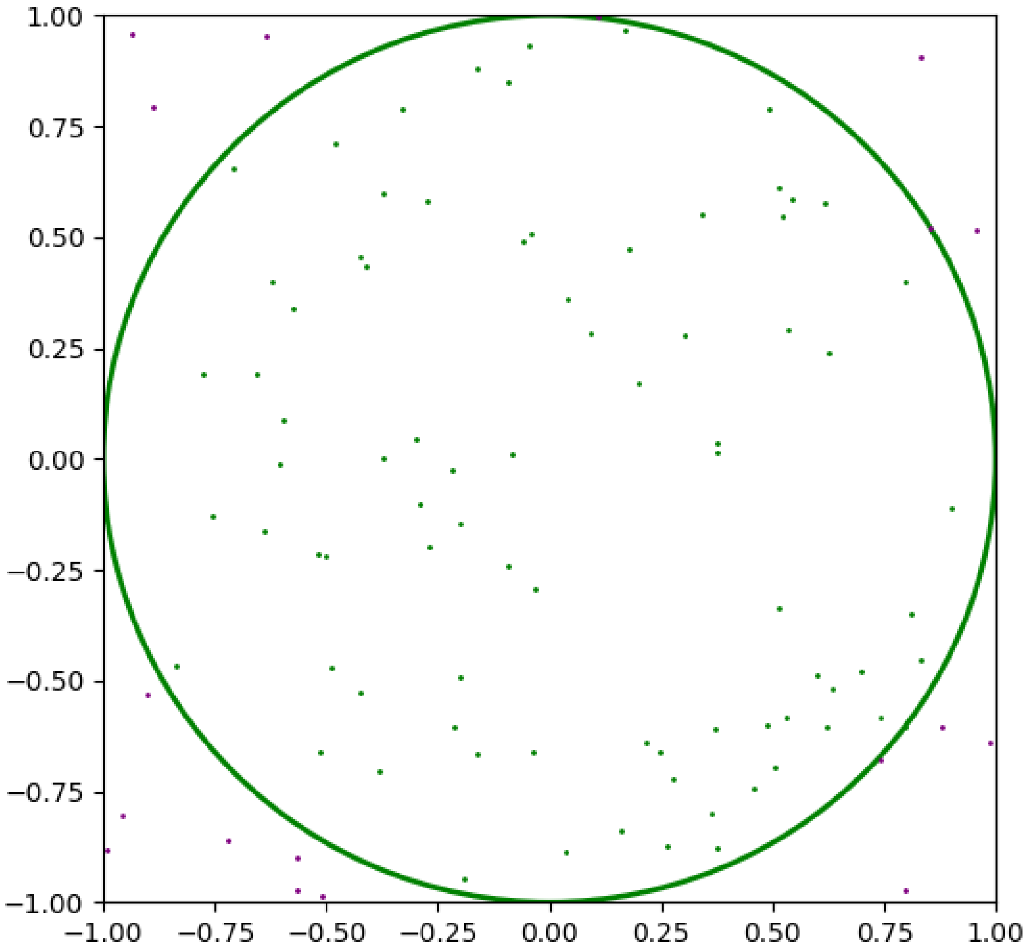
Question 1

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches
# add plot of unit circle
    fig, ax = plt.subplots(figsize=(6, 6))
    circle = patches.Circle((0, 0), 1, edgecolor='green',
    facecolor='none', linewidth=2)
    ax.add_patch(circle)
    ax.set_xlim(-1, 1)
    ax.set_ylim(-1, 1)
# generate random points and plot them
    rng = np.random.default_rng(714)
    N = 100
    x = rng.uniform(-1, 1, N)
    y = rng.uniform(-1, 1, N)
    inside_circle = x**2 + y**2 <= 1
    ax.scatter(x[inside_circle], y[inside_circle], color='yellow', s=1)
    ax.scatter(x[~inside_circle], y[~inside_circle], color='blue', s=1)
# estimate pi
    tau = inside_circle.sum()
    pi_est = 4 * tau / N
# error term
    error = abs(np.pi - pi_est)
    (pi_est, error)
```

N	π_N	e_N
100	3.28	0.1384
1,000	3.052	0.0895
10,000	3.118	0.0235
100,000	3.13876	0.0028



Question 2

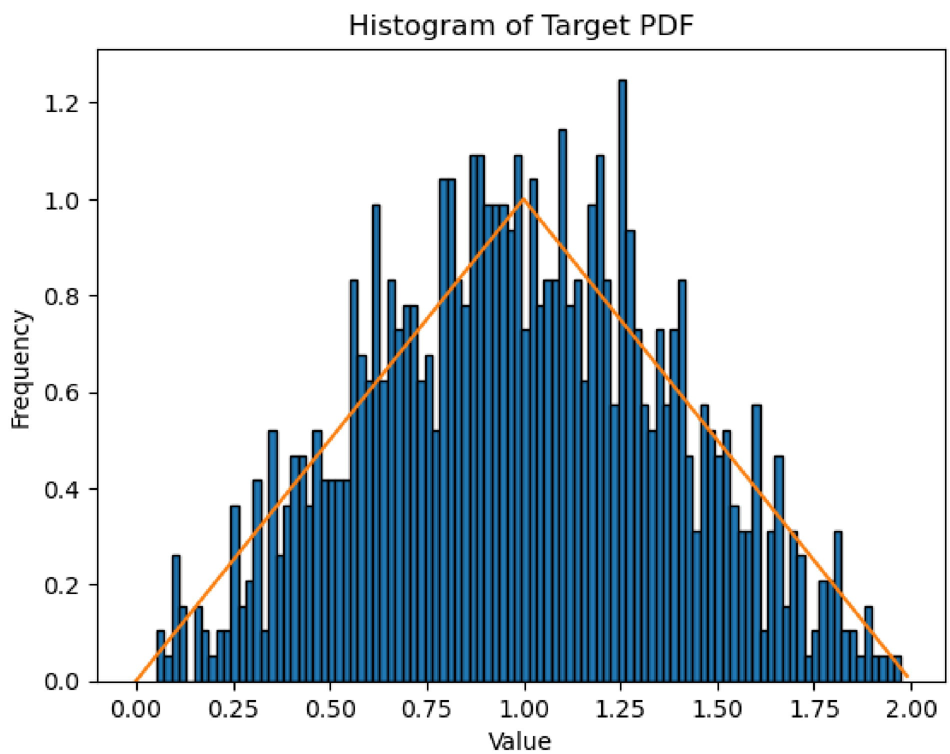
a. First we compute the CDF from the given PDF.

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \int_0^1 x dx & \text{if } 0 \le x \le 1 \\ \int_0^1 x dx + \int_1^2 (2-x) dx & \text{if } 1 < x \le 2 \\ 1 & \text{if } x > 2 \end{cases}$$

$$= \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{2}x^2 & \text{if } 0 \le x \le 1 \\ -1 + 2x - \frac{1}{2}x^2 & \text{if } 1 < x \le 2 \\ 1 & \text{if } x > 2 \end{cases}$$

Now solve $F(x) = u$

$$X = \begin{cases} \sqrt{2u} & \text{if } 0 \le u \le \frac{1}{2} \\ 2 - \sqrt{2 - 2u} & \text{if } \frac{1}{2} < x \le 1 \end{cases}$$



b.

Question 3